



UC San Diego
Electrical and Computer Engineering
JACOBS SCHOOL OF ENGINEERING

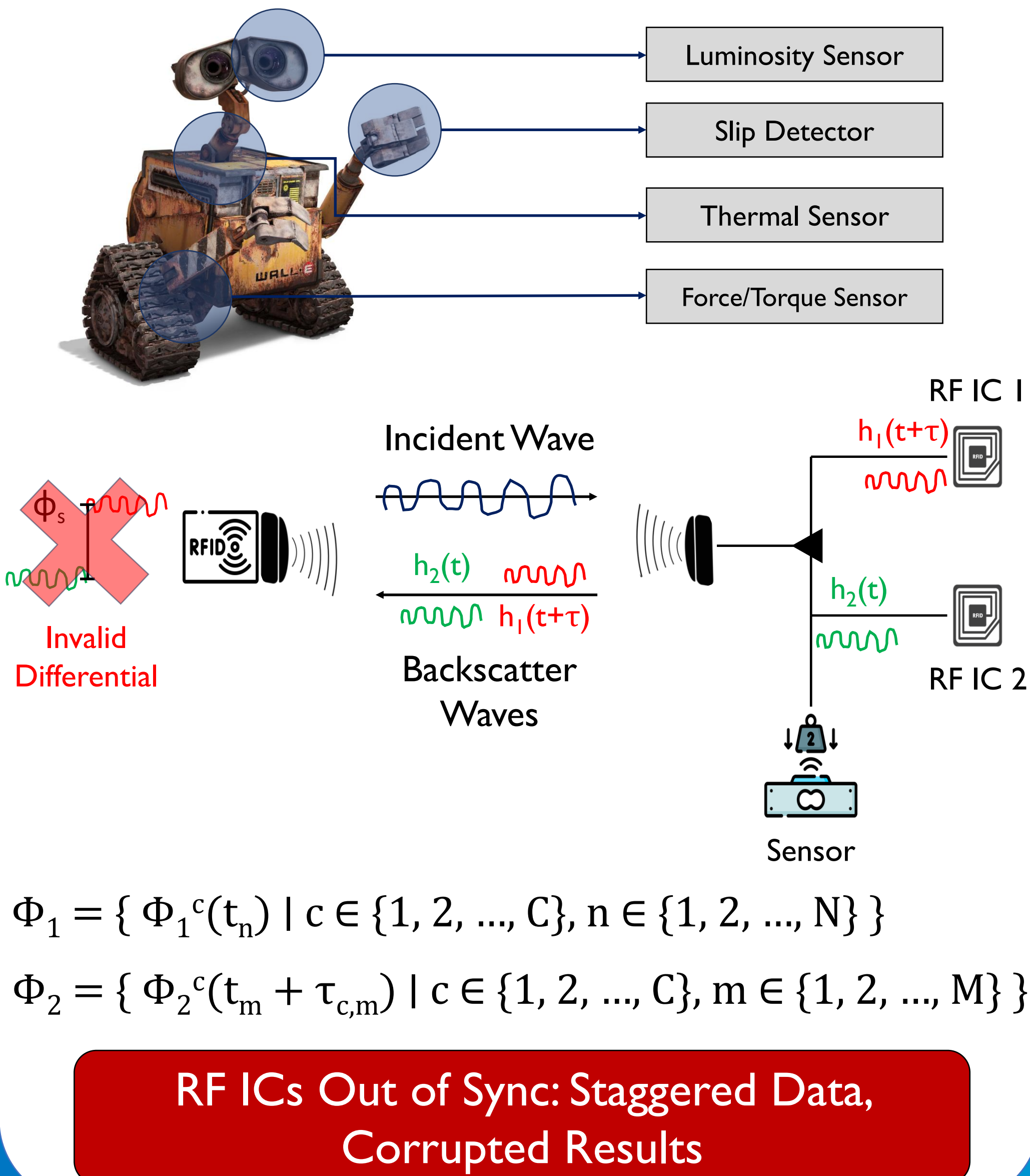
SenSync

Accurate and Real-Time Passive Sensing

Ishan Bansal, Nagarjun Bhat, Agrim Gupta, Harine Govindarajan, Dinesh Bharadia

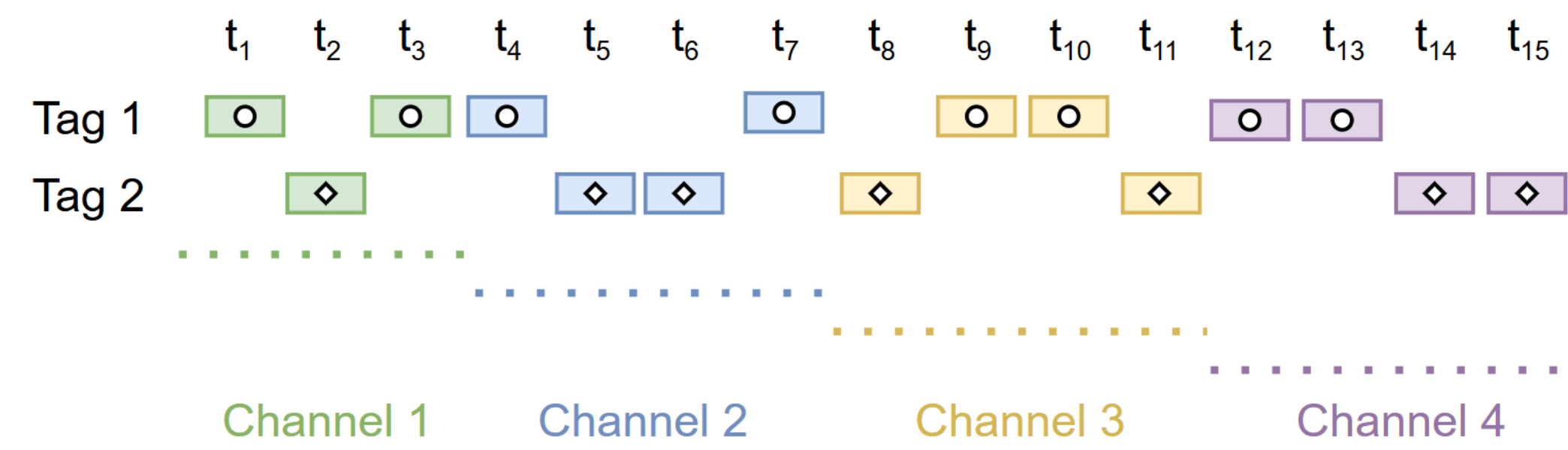


Differential RFID Sensing

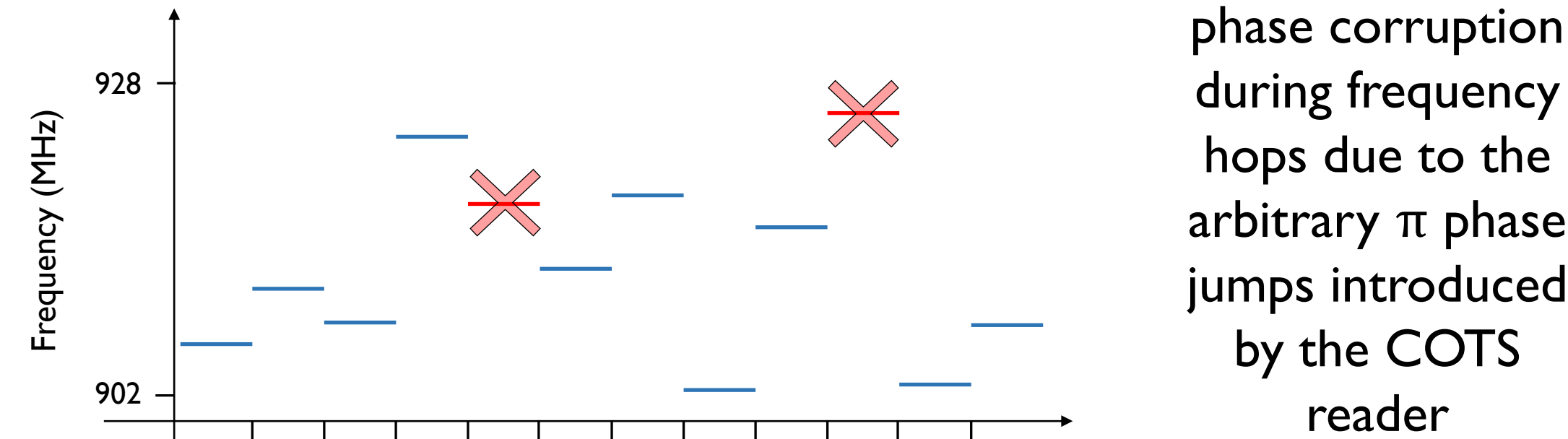


Caveats with COTS Readers

Sequential Data Processing



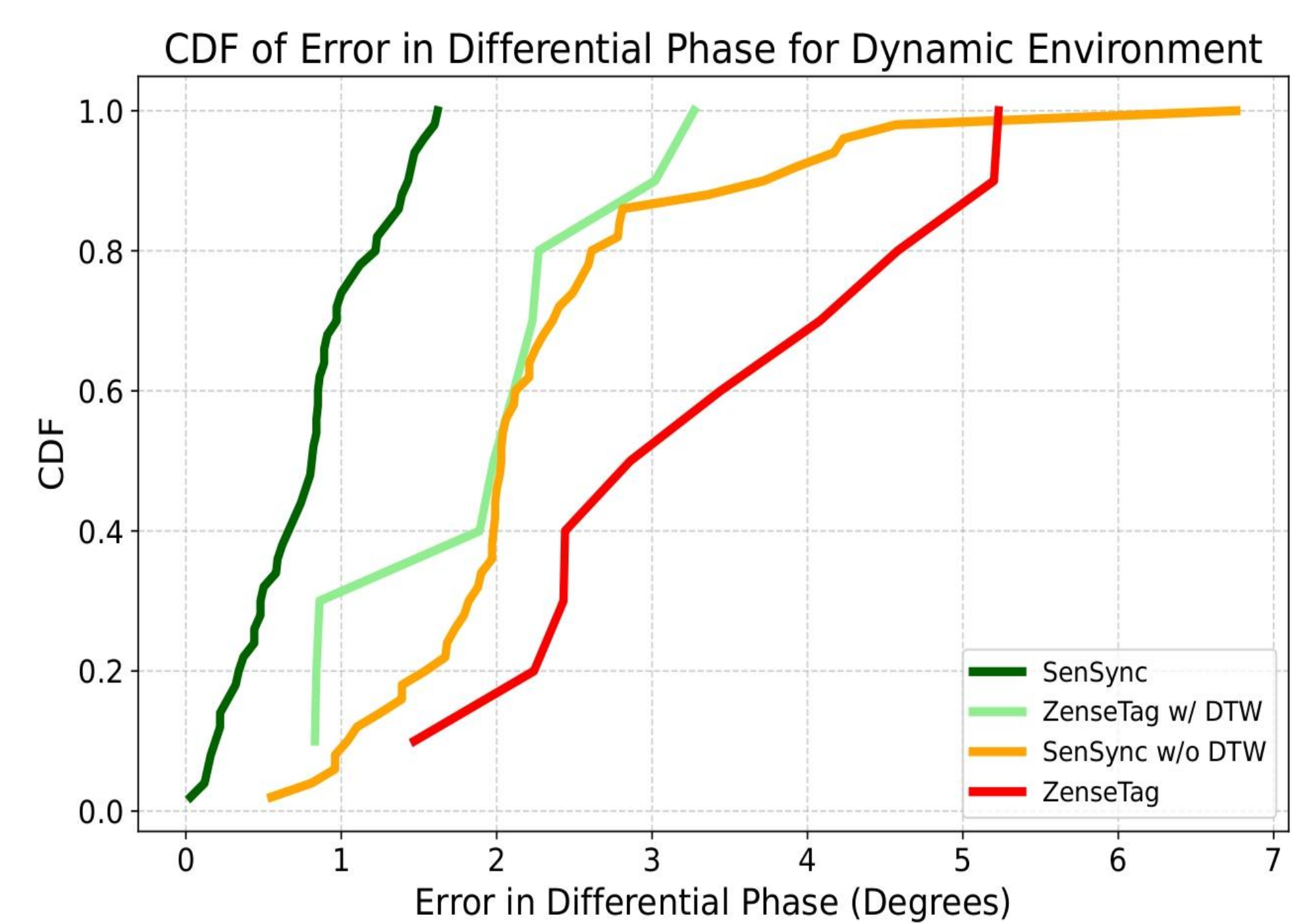
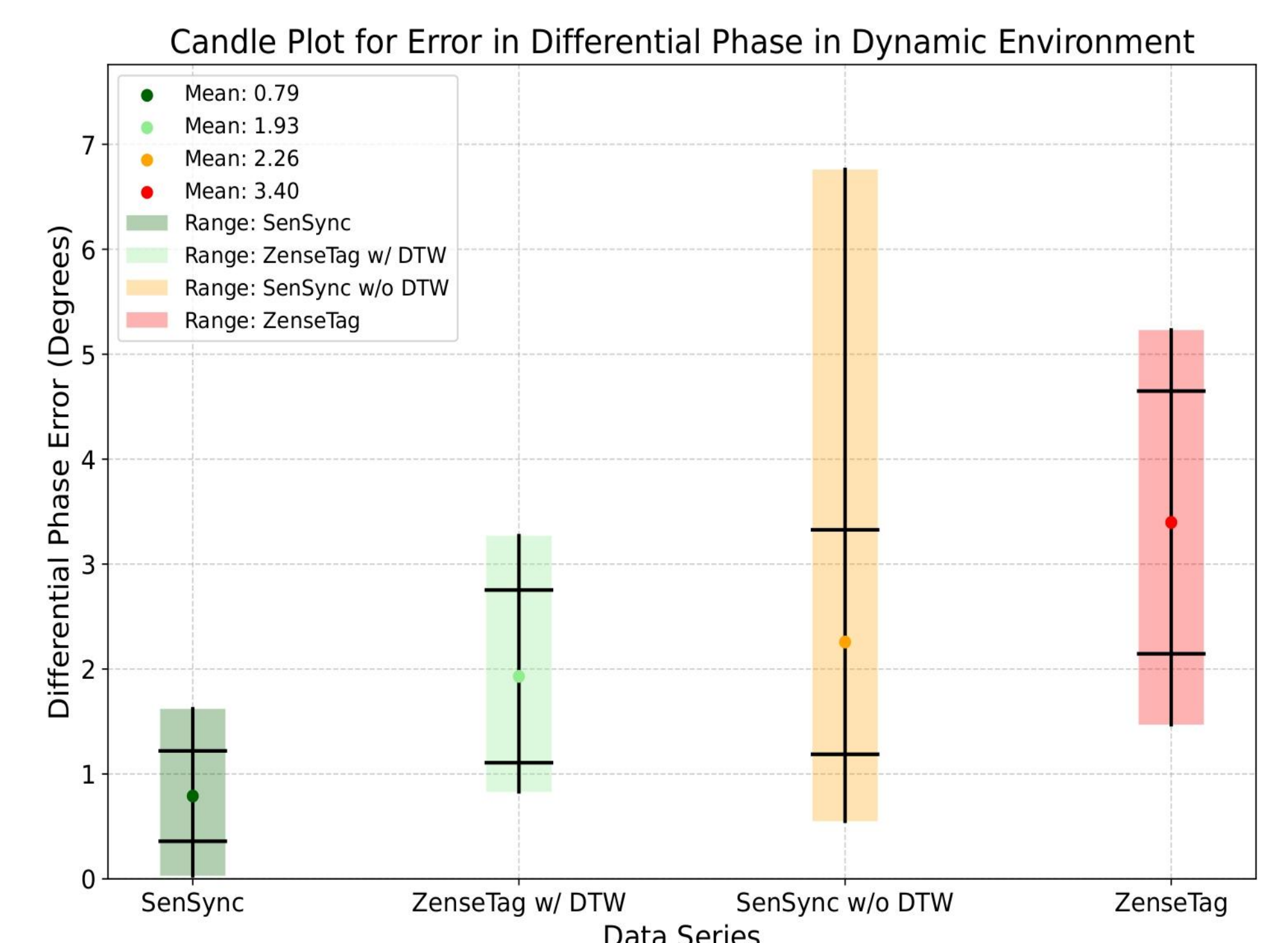
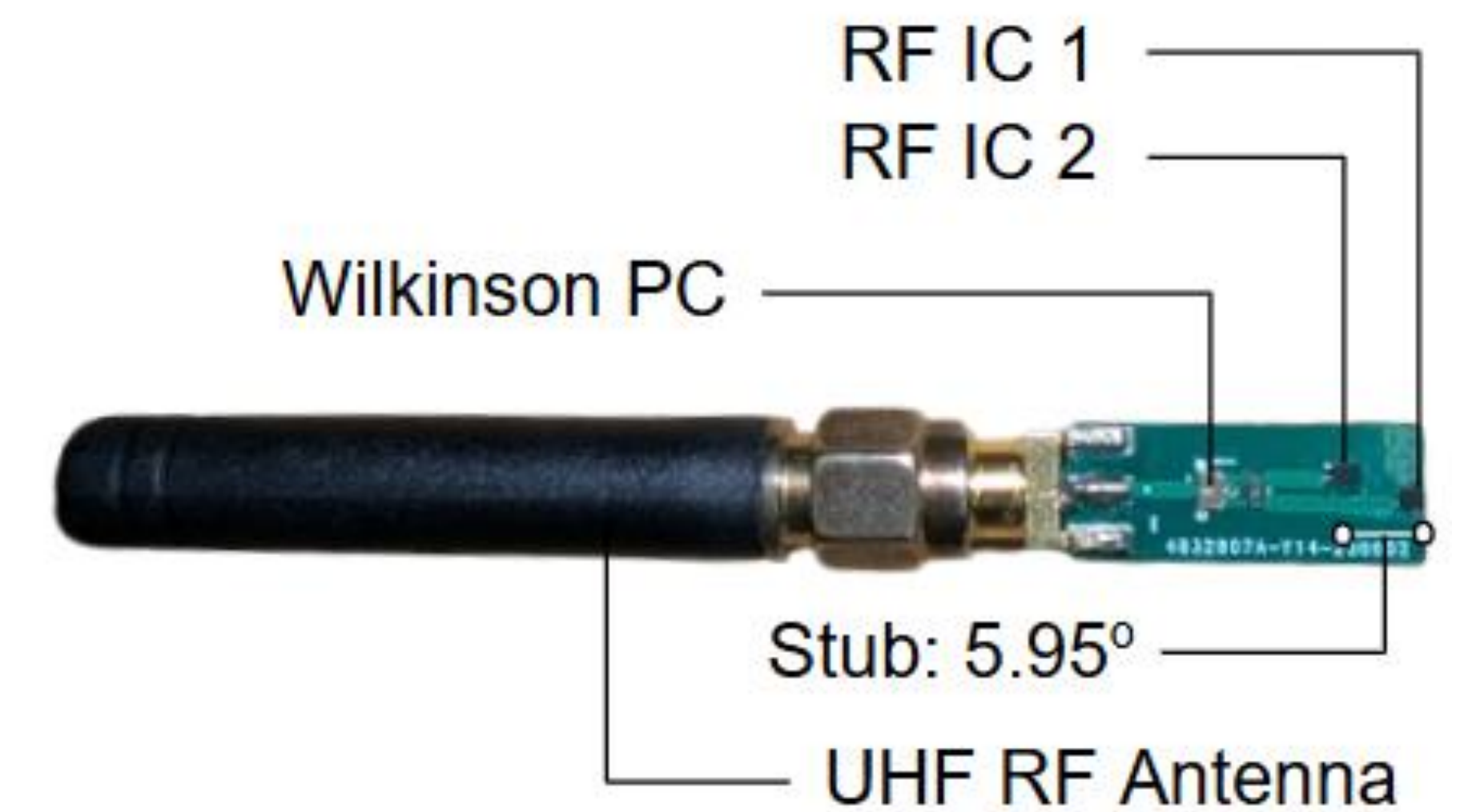
Frequency Hopping



Sensor Data: Flicker of Truth?
Data Stream Retrieval can Deceive

The Proof of the Pudding is in the Eating!

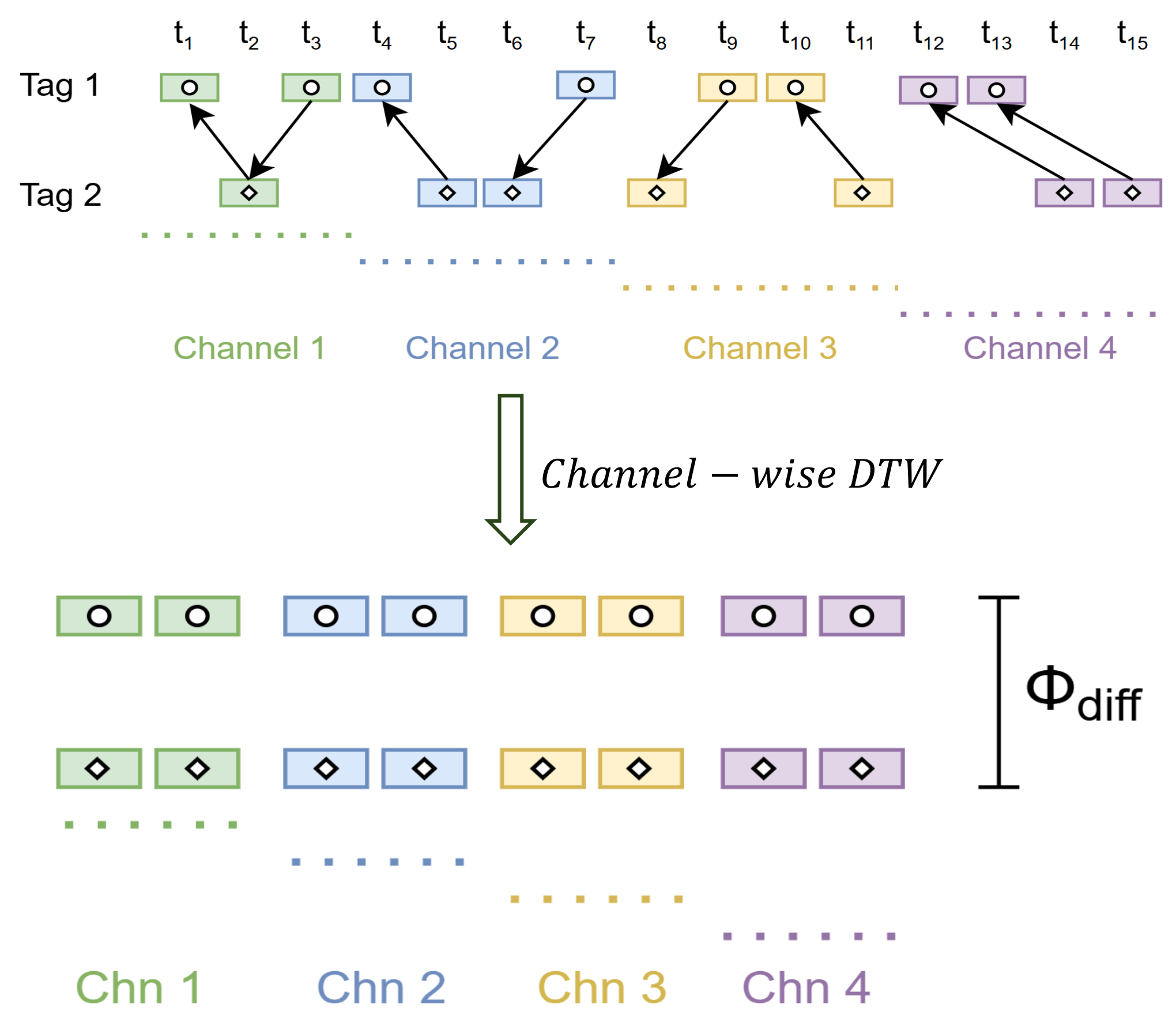
Simulatory Stubbed Tag (SST)



When Less is More: SenSync is 4x more accurate at sensing capabilities than State-of-the-Art Sensing Systems

Solving the Data Puzzle

Secret Sauce – Dynamic Time Warping

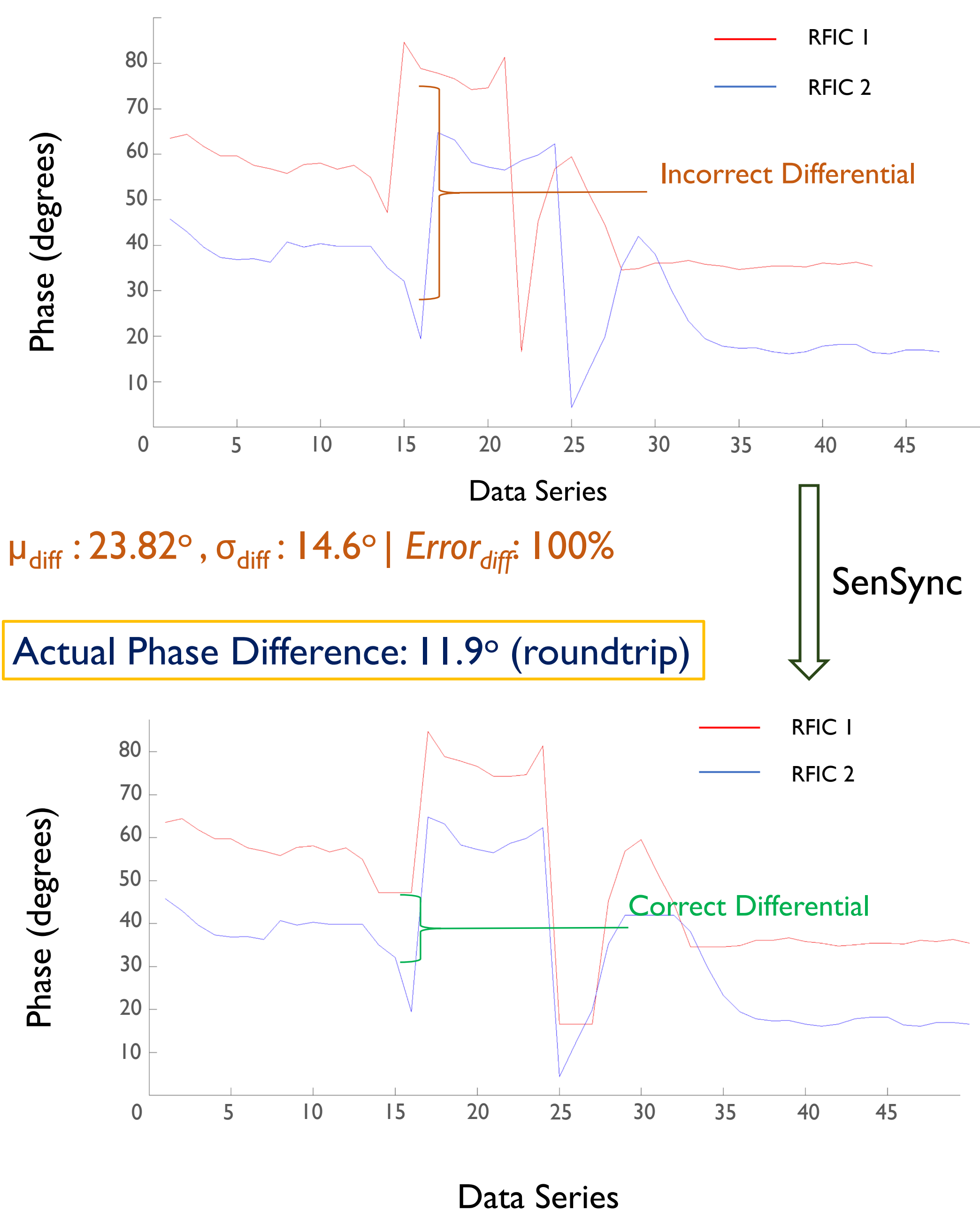


$$\Phi_{diff}^c(t_i) = | \Phi_1^c(t_i) - \Phi_2^c(t_i) | \forall t_i \in W, c \in \{1, 2, 3, 4\}$$

$$\Phi_{diff} = (\bar{\Phi}_{diff}^1(t_i) + \bar{\Phi}_{diff}^2(t_i) + \bar{\Phi}_{diff}^3(t_i) + \bar{\Phi}_{diff}^4(t_i)) / \sum_{i=1}^N i$$

Reliability Redefined: Transforming Chaotic RFID signals into Clear Sensing data

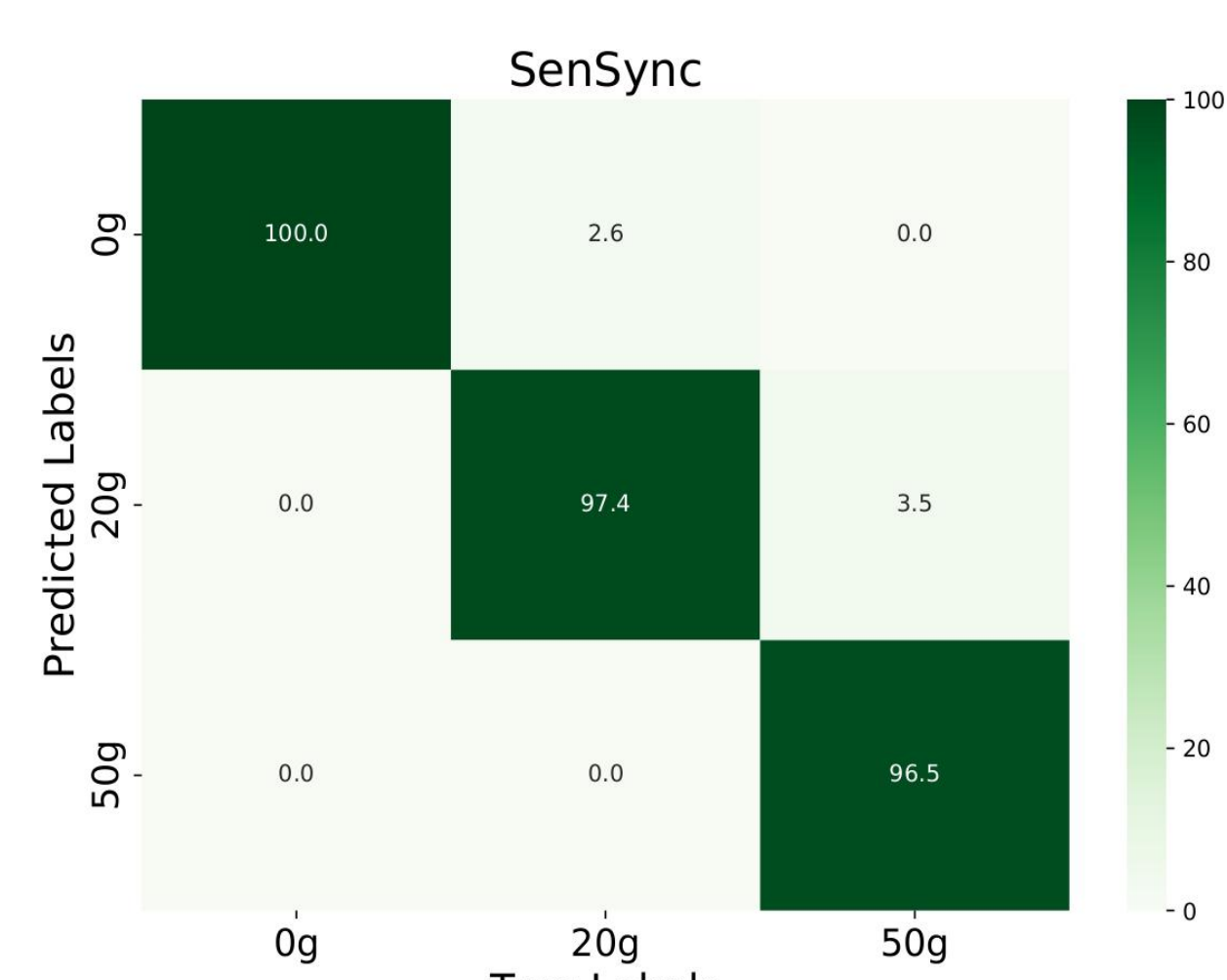
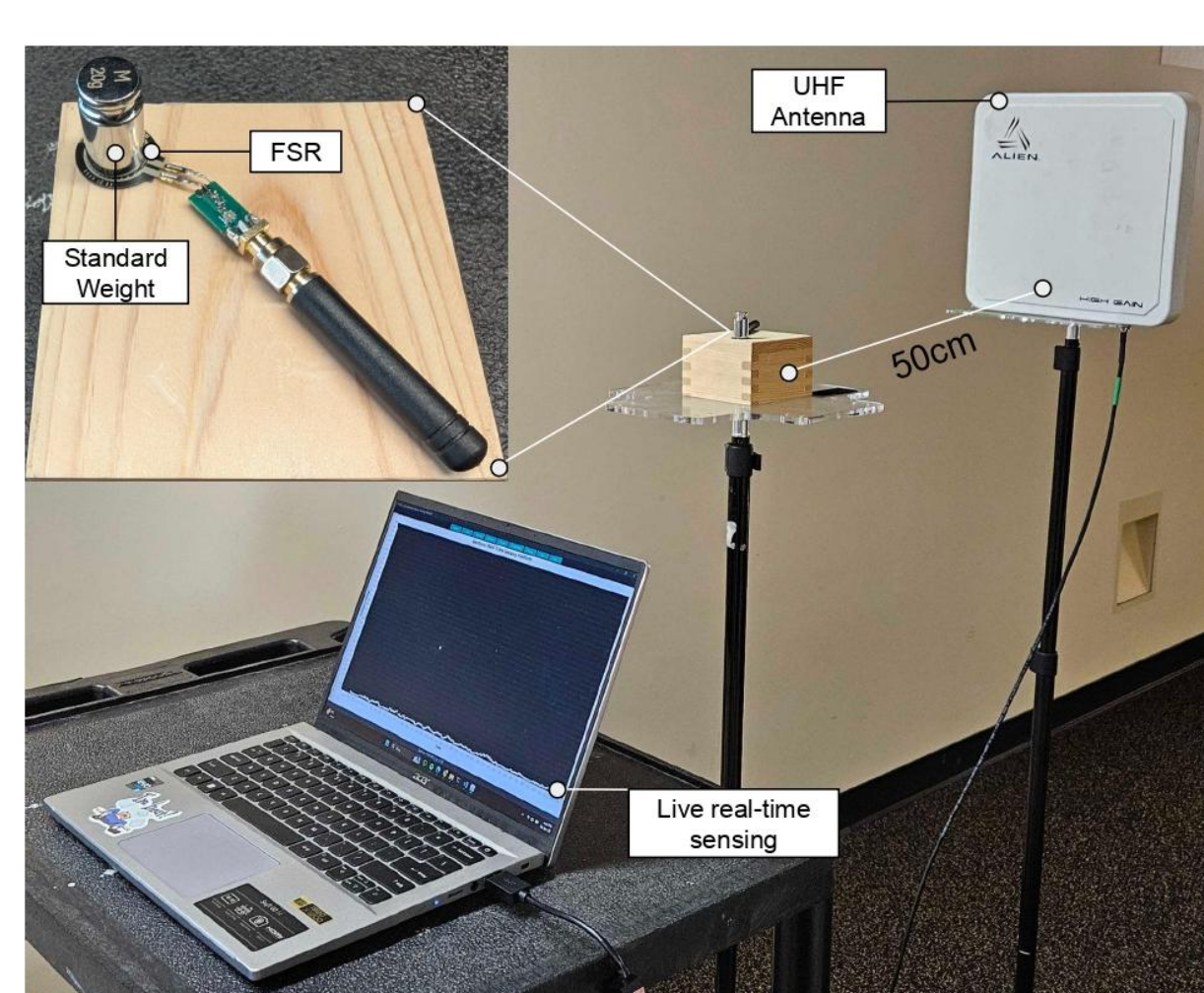
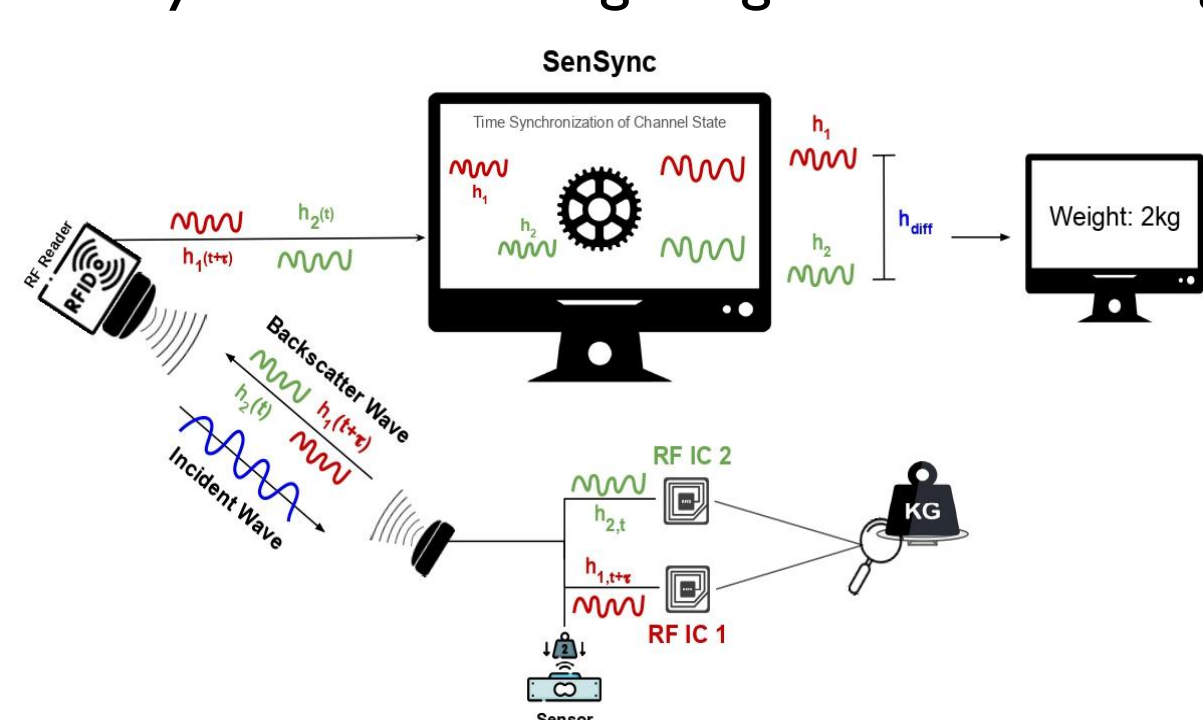
Reality Realigned



Sequence Alignment unlocks Meaningful Phase Insights

The Clear Winner

- SenSync** achieves:
- 4x better accuracy than State-of-Art techniques
 - 5x improvement in sensing latency
 - 8x improvement in sensing data throughput
 - 97% accuracy when detecting weights as low as 20g and 50g



Summary

SenSync is an innovative algorithm that significantly improves RFID-based differential sensing. It addresses temporal misalignment and phase ambiguity issues, achieving **5x faster** sensory resolution and **8x higher throughput** compared to existing methods. SenSync demonstrates superior accuracy and robustness in dynamic environments.

